Supplementary Material for "When Are Triangles Invisible? Fundamental Limits of Higher-Order Structure Identifiability from Edge Flows"

Weiping Yan
University of Minnesota, Twin Cities

This supplement extends the main paper in three directions: **S1** Fano-type converses for *joint* support recovery (including a signal-distribution-agnostic bound and a sub-Gaussian achievability counterpart); **S2** identifiability under *partial edge observation*; **S3** *plug-in* estimation of the variance parameters and a fully adaptive detector. All statements are validated numerically by the released test-suite; Section S6 maps every figure to its script. Notation follows the main paper: N snapshots, candidate set $\mathcal{T}$ with $p = |\mathcal{T}|$, latent support $\mathcal{S}$ with $|\mathcal{S}| = k$, curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$, per-coordinate null/alternative variances $v_0 = 3\sigma_n^2$, $v_1 = v_0(1+\rho)$. Throughout S1–S2 the candidates are pairwise edge-disjoint, so the curl coordinates are independent given $\mathcal{S}$ and the triangle Gram matrix is $\boldsymbol{G} = 3\boldsymbol{I}$.

S1 Fano-type converses for joint support recovery

The main paper's Theorem 1 is a *single-triangle* converse: it bounds the error of deciding one triangle's activity and yields the floor $\rho^\star(N) = \Theta(\sqrt{\log(1/\delta)/N})$. Recovering the whole support $\mathcal{S}$ among $\binom{p}{k}$ possibilities is harder. The next two theorems quantify by how much.

We first record a reduction used by both proofs.

Lemma S1.1 (Sufficiency of the curl coordinates)**.** *Consider the model of the main paper (Eq. (2)) with isotropic Gaussian noise $\boldsymbol{n}_t \sim \mathcal{N}(0, \sigma_n^2\boldsymbol{I})$, mutually independent $(\boldsymbol{a}_t, \boldsymbol{y}_t, \boldsymbol{h}_t, \boldsymbol{n}_t)$, and* arbitrary *laws for $(\boldsymbol{a}_t, \boldsymbol{y}_t, \boldsymbol{h}_t)$ (Gaussianity is needed only for the noise). Then*

$$I(\mathcal{S}; \{\boldsymbol{f}_t\}_{t=1}^N) \;=\; I(\mathcal{S}; \{\boldsymbol{c}_t\}_{t=1}^N), \qquad \boldsymbol{c}_t = \boldsymbol{B}_2^\top \boldsymbol{f}_t.$$

Proof. Decompose $\boldsymbol{f}_t$ orthogonally into its gradient, curl, and harmonic components. Because the isotropic Gaussian noise has independent projections onto orthogonal subspaces, and the gradient (resp. harmonic, curl) component of $\boldsymbol{f}_t$ equals $\boldsymbol{B}_1^\top \boldsymbol{a}_t$ (resp. $\boldsymbol{h}_t$, $\boldsymbol{B}_{2,\mathcal{S}}\boldsymbol{y}_t$) plus the corresponding independent noise projection, the three components are mutually independent, and only the curl component depends on $\mathcal{S}$. Hence the gradient and harmonic components are independent of $(\mathcal{S}, \text{curl component})$ and carry zero information about $\mathcal{S}$. Finally, on edge-disjoint candidates $\boldsymbol{B}_2$ has full column rank, so $\boldsymbol{c}_t$ is a bijective linear image of the curl component. $\square$

Theorem S1.2 (Gaussian Fano converse)**.** *Let $\mathcal{S}$ be uniform on the $\binom{p}{k}$ supports of size k, with Gaussian signals and noise and edge-disjoint candidates. If an estimator achieves $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \le \varepsilon$, then*

$$N \;\ge\; \frac{(1-\varepsilon)\log\binom{p}{k} - \log 2}{k\,\mathrm{KL}_1(\rho)}, \qquad \mathrm{KL}_1(\rho) = \tfrac{1}{2}\big(\rho - \log(1+\rho)\big) \;\le\; \tfrac{\rho^2}{4}. \tag{1}$$

In particular $\rho^2 N \gtrsim 4(1-\varepsilon)\log(p/k)$*: joint recovery pays a* $\log\binom{p}{k}$ *factor where the single-triangle test pays* $\log(1/\delta)$*.*

Proof. By Lemma S1.1 and Fano's inequality, $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \geq 1 - (I(\mathcal{S}; \boldsymbol{C}^N) + \log 2)/\log\binom{p}{k}$ with $\boldsymbol{C}^N = \{\boldsymbol{c}_t\}$. Bound the mutual information against the all-inactive reference law P_0: $I(\mathcal{S}; \boldsymbol{C}^N) \leq \frac{1}{\binom{p}{k}} \sum_{\mathcal{S}} \mathrm{KL}(P_{\mathcal{S}}^{\otimes N} \| P_0^{\otimes N}) = N\,\mathrm{KL}(P_{\mathcal{S}} \| P_0)$ for any fixed $\mathcal{S}$ by symmetry. Given $\mathcal{S}$, the coordinates of $\boldsymbol{c}$ are independent, equal in law to $\mathcal{N}(0, v_1)$ on $\mathcal{S}$ and $\mathcal{N}(0, v_0)$ off $\mathcal{S}$, and the reference has all coordinates $\mathcal{N}(0, v_0)$; hence $\mathrm{KL}(P_{\mathcal{S}} \| P_0) = k\,\mathrm{KL}(\mathcal{N}(0, v_1) \| \mathcal{N}(0, v_0)) = \frac{k}{2}\big(\frac{v_1}{v_0} - 1 - \log\frac{v_1}{v_0}\big) = k\,\mathrm{KL}_1(\rho)$. Rearranging Fano gives (1); the inequality $\mathrm{KL}_1 \leq \rho^2/4$ is $\log(1+\rho) \geq \rho - \rho^2/2$. □

Theorem S1.3 (Signal-agnostic Fano converse)**.** *Keep the noise Gaussian, but let the active triangle signals* $y_{\tau,t}$ *be i.i.d. from an* arbitrary *zero-mean law with variance* σ_c^2*. Then any estimator with error at most* ε *requires*

$$N \;\geq\; \frac{(1-\varepsilon)\log\binom{p}{k} - \log 2}{\frac{p}{2}\,\log\big(1 + \rho k/p\big)}. \tag{2}$$

Proof. Lemma S1.1 applies verbatim (its proof used only independence and Gaussianity of the *noise*). With conditionally i.i.d. snapshots, $I(\mathcal{S}; \boldsymbol{C}^N) \leq N\, I(\mathcal{S}; \boldsymbol{c}_1)$, and with conditionally independent coordinates, subadditivity of entropy gives $I(\mathcal{S}; \boldsymbol{c}) \leq \sum_\tau \big[h(c_\tau) - h(c_\tau \mid \mathcal{S})\big]$. Under the uniform prior each coordinate is active with probability k/p, so $\mathrm{Var}(c_\tau) = v_0(1 + \rho k/p)$ and the Gaussian maximum-entropy bound gives $h(c_\tau) \leq \frac{1}{2}\log(2\pi e\, v_0(1 + \rho k/p))$. Conditionally on $\mathcal{S}$, c_τ is an independent signal plus $\mathcal{N}(0, v_0)$ noise, so $h(c_\tau \mid \mathcal{S}) \geq \frac{1}{2}\log(2\pi e\, v_0)$ (adding an independent variable cannot decrease entropy). Summing over the p coordinates yields $I(\mathcal{S}; \boldsymbol{c}) \leq \frac{p}{2}\log(1 + \rho k/p)$ and Fano concludes. □

Remark S1.4 (How the two converses compare)**.** *Both bounds are valid for Gaussian signals, so one may take their pointwise maximum. For sparse supports* ($k \ll p$) *and small* ρ *the Gaussian bound is stronger by a factor* $\asymp 1/\rho$ *(its denominator is* $k\mathrm{KL}_1 \asymp k\rho^2/4$ *versus* $\asymp k\rho/2$*). In the dense regime* ($k \asymp p$) *the agnostic budget grows only logarithmically in* ρ*, so it* overtakes *the Gaussian-KL bound at high SNR (Fig. 1B). Figure 1A shows the* $\log p$ *effect: at the scale of the paper's Fig. 1* ($p = 8$) *the joint-recovery floor sits well below the per-triangle Chernoff floor, while at* $p = 10^4$ *it narrows the gap to a* $\approx 2\times$ *factor (the curves are not directly comparable in level: Fano is normalized at error* $1/2$*, the Chernoff floor at* $\delta = 0.05$*).*

Proposition S1.5 (Sub-Gaussian achievability)**.** *Suppose the curl-coordinate noise and signals are sub-Gaussian with* ψ_2*-norms bounded by* K *times their standard deviations. Then the energy detector with threshold* $\gamma = v_0(1 + \rho/2)$ *recovers* $\mathcal{S}$ *exactly with probability* $\geq 1 - \delta$ *once*

$$N \;\geq\; \frac{C\,K^4}{\min(\rho, \rho^2)/(1+\rho)^2}\,\log\frac{2p}{\delta},$$

for a universal constant C*: the same* $\log p/\rho^2$ *scaling as the Gaussian theory, with constants degraded by* K^4*.*

Proof sketch. Each $c_{\tau,t}^2 - \mathbb{E}c_{\tau,t}^2$ is sub-exponential with ψ_1-norm $\lesssim K^2 v_i$. Bernstein's inequality [1, Thm. 2.8.1] gives, for the empirical energy $\hat{E}_\tau = \frac{1}{N}\sum_t c_{\tau,t}^2$,

$$\Pr\big[|\hat{E}_\tau - v_i| \geq t\big] \;\leq\; 2\exp\Big(-cN\min\big(t^2/(K^4 v_i^2),\; t/(K^2 v_i)\big)\Big).$$

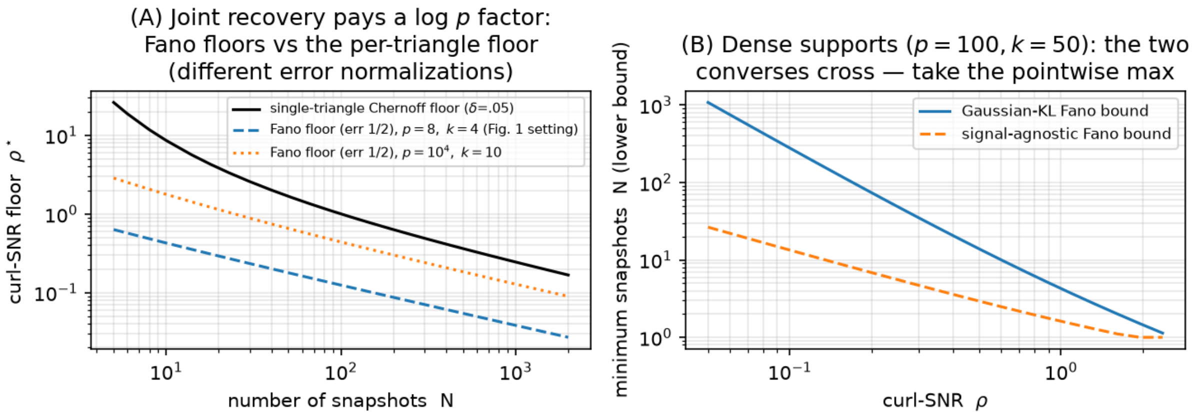

Figure 1: (A) Fano floors (Thm. S1.2, inverted in ρ at error 1/2) against the single-triangle Chernoff floor of the main paper. (B) Dense-support regime: the Gaussian-KL and signal-agnostic bounds cross. Script: `experiments/run_fano.py`.

The two hypotheses are separated by $v_1 - v_0 = v_0\rho$; taking $t = v_0\rho/2$ and a union bound over the p candidates yields the claim. $\square$

S2 Partial edge observation

Suppose only a subset $\mathcal{E}_{\text{obs}} \subseteq \mathcal{E}$ of edges is observed — each edge independently with probability q (Bernoulli sampling). Write $\boldsymbol{f}_t^{\text{obs}}$ for the observed sub-vector.

Proposition S2.1 (Observability for curl statistics)**.** *The curl statistic $c_\tau = \boldsymbol{B}_2[:,\tau]^\top \boldsymbol{f}$ of candidate τ is computable from $\boldsymbol{f}^{\text{obs}}$ iff* all three *edges of τ are observed. Under Bernoulli(q) sampling each triangle is observable with probability q^3, independently across edge-disjoint candidates.*

An unobservable *active* triangle is an automatic miss; an unobservable *inactive* one is an automatic correct rejection (the detector defaults to "inactive"). Combining with the exact per-triangle marginal laws (main paper, Thm. 2) gives a closed form on edge-disjoint geometries:

Corollary S2.2 (Exact recovery under edge sampling)**.** *With edge-disjoint candidates, the centered whitened detector at budget N achieves*

$$\Pr[\hat{\mathcal{S}} = \mathcal{S}] = \left[q^3 \left(1 - P_{\text{miss}}\right)\right]^k \left[1 - q^3 P_{\text{fa}}\right]^{p-k}, \tag{3}$$

where $P_{\text{miss}}, P_{\text{fa}}$ are the χ^2_{N-1} error probabilities of the fully observed test.

Proof. Independence of survival (Prop. S2.1) and of the whitened scores ($\boldsymbol{G} = 3\boldsymbol{I}$). An active triangle is recovered iff it survives (prob. q^3) *and* is detected ($1 - P_{\text{miss}}$); an inactive one is mislabeled iff it survives *and* false-alarms ($q^3 P_{\text{fa}}$). $\square$

The q^{3k} factor is brutal: partial sampling multiplies the fully observed recovery probability by $q^{3k} = q^{54}$ at $k = 18$ active triangles — a factor ≈ 0.06 at $q = 0.95$ and ≈ 0.003 at $q = 0.9$ — and the Monte-Carlo cliff in Fig. 2B follows the closed form within sampling error. The restricted curve (recovery on the observable candidates only) stays at ≈ 1 for all q: the bottleneck is purely coverage, not detection.

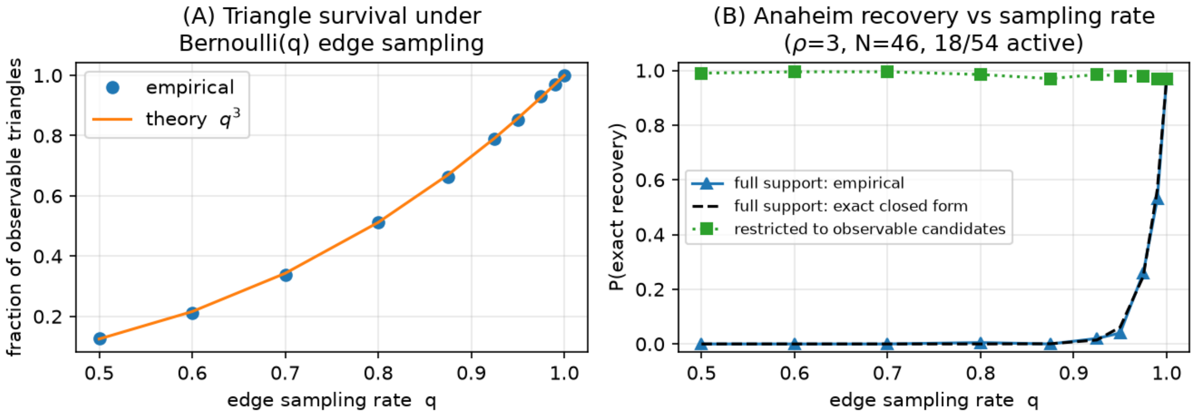

Figure 2: Partial edge observation on Anaheim (real geometry and equilibrium flow background, $\rho = 3$, $N = 46$, 18/54 active). (A) Triangle survival follows q^3. (B) Full-support recovery collapses along the exact closed form (3); recovery restricted to observable candidates stays ≈ 1. Script: `experiments/run_partial_sampling.py`.

Remark S2.3 (Scope)**.** *Proposition S2.1 is an* operational *statement about detectors built on curl statistics — the class this paper analyzes. A triangle with two observed edges still injects signal into those edges' flows, but that signal is entangled with the unknown gradient nuisance once the exact curl cancellation is unavailable; whether (and at what cost) it can be exploited, e.g. by marginalizing the gradient over the observed sub-graph, is an open question we do not claim to settle. Low-rank flow completion under sampling (cf. graph sampling theory [2]) is the natural attack.*

S3 Plug-in variance estimation and the adaptive detector

The paper's detectors take (σ_c, σ_n) as known. We now estimate both from the same data. Let $u_\tau = \mathrm{score}_\tau / (\boldsymbol{G}^{-1})_{\tau\tau}$ denote the normalized whitened scores, so that for inactive τ, $N u_\tau / \sigma_n^2 \sim \chi_d^2$ ($d = N$ or $N - 1$ if centered), while active scores are stochastically larger.

Estimators.

$$\hat{\sigma}_n^2 = \frac{N\,\mathrm{med}_\tau(u_\tau)}{\mathrm{med}(\chi_d^2)}, \qquad \text{(median, robust to} <50\% \text{ contamination)} \tag{4}$$

$$\hat{\sigma}_c^2 = \Big(\tfrac{1}{|\hat{\mathcal{A}}|}\textstyle\sum_{\tau\in\hat{\mathcal{A}}}\big[\mathrm{score}_\tau - \hat{v}_{0,\tau}\big]\Big)_+, \qquad \hat{\mathcal{A}} = \text{Bonferroni screen at level } \alpha, \tag{5}$$

followed by one refinement pass: after a first detection, σ_n is re-fit by the mean of the *detected* off-support scores and σ_c by the on-support excess, and the detector is re-run once. Because the refit set is selected by the same data, the refit is conditionally biased (missed active triangles inflate $\hat{\sigma}_n$ — measured $\approx +5\%$ at $N = 10$ on the strip benchmark, vanishing for $N \geq 20$); its accuracy is established empirically (Fig. 3A), not by an unbiasedness argument, and Prop. S3.2 applies conditionally on the deterministic error bound ϵ.

Lemma S3.1 (Median consistency envelope)**.** *Assume the candidates are pairwise edge-disjoint, so that the inactive normalized scores are i.i.d. $\sigma_n^2\chi_d^2/N$ (the DKW step below requires independence; for correlated scores the envelope is validated only empirically). Let a fraction $\pi < 1/2$ of the p candidates be active and let $\epsilon_p = \sqrt{\log(2/\delta)/(2(1-\pi)p)}$ satisfy $\pi + \epsilon_p < 1/2$. Then with probability at least $1-\delta$,*

$$q_d\Big(\tfrac{1/2-\pi-\epsilon_p}{1-\pi}\Big) \;\le\; \frac{N\,\mathrm{med}_\tau(u_\tau)}{\sigma_n^2} \;\le\; q_d\Big(\tfrac{1/2}{1-\pi} + \tfrac{\epsilon_p}{1-\pi}\Big),$$

where $q_d(\cdot)$ is the χ_d^2 quantile function. Consequently $\hat\sigma_n^2/\sigma_n^2 \in [q_d(\frac{1/2-\pi-\epsilon_p}{1-\pi}), q_d(\frac{1/2+\epsilon_p}{1-\pi})]/q_d(1/2)$ — an explicit, numerically computable envelope that shrinks to $\{1\}$ as $p, d \to \infty$ whenever $\pi < 1/2$.

Proof. Active scores stochastically dominate inactive ones, so the sample median of all p scores is sandwiched between order statistics of the $(1-\pi)p$ inactive scores: at least a $(1/2-\pi)/(1-\pi)$-fraction and at most a $(1/2)/(1-\pi)$-fraction of the inactive sample lies below it. The Dvoretzky–Kiefer–Wolfowitz inequality applied to the inactive empirical CDF converts these fractions into population χ_d^2 quantiles up to ϵ_p, with probability $1-\delta$. □

Proposition S3.2 (Plug-in perturbation)**.** *Let the plug-in variances satisfy $\hat v_{i,\tau} = (1+e_{i,\tau})v_{i,\tau}$ with $|e_{i,\tau}| \le \epsilon < 1/2$. Then each per-triangle error probability of the adaptive detector exceeds its known-parameter counterpart by at most $L_d\,\epsilon$, where $L_d < \infty$ is a Lipschitz constant of the χ_d^2 CDFs at the optimal threshold, and the exact-recovery probability drops by at most $p\,L_d\,\epsilon$ (union bound).*

Proof sketch. The Bayes threshold $\gamma(v_0, v_1)$ of the two-variance test is a smooth function of (v_0, v_1) (main paper, Prop. 1), so $|\hat\gamma - \gamma| \lesssim \epsilon\gamma$; the error probabilities are integrals of bounded χ_d^2 densities over the threshold shift. □

Empirically (Fig. 3) both estimates contract at the $1/\sqrt{N}$ rate and the fully adaptive detector's recovery curve is indistinguishable from the known-parameter detector for $N \gtrsim 30$ on the confusable strip benchmark; the residual gap at $N \le 20$ (at most 0.06 after refinement) is the price of estimating a variance from nine medians.

S4 First- vs. second-order identifiability: proofs

This section proves the main paper's Lemma 2 (lifted spark) and Theorem 3 (first- vs. second-order dichotomy), and records the exhaustive numerical verifications. Here the candidate set is arbitrary (edge-sharing allowed, $\boldsymbol{B}_2$ possibly rank-deficient); $\rho_2 = \sigma_c^2/\sigma_n^2$ denotes the second-order SNR (the per-triangle curl-SNR of the main text is $\rho = 3\rho_2$).

S4.1 The lifted spark lemma

Lemma S4.1 (Lifted spark)**.** *Let $\mathcal{T}$ be any set of distinct 3-cliques of a simple graph, with signatures $\boldsymbol{b}_\tau \in \{0, \pm 1\}^{|\mathcal{E}|}$ (columns of $\boldsymbol{B}_2$). The matrices $\{\boldsymbol{b}_\tau\boldsymbol{b}_\tau^\top\}_{\tau\in\mathcal{T}}$ are linearly independent.*

Proof. Fix $\tau = (i, j, k)$ and its edge pair $e = (i, j)$, $e' = (j, k)$. A pair of distinct edges determines at most one triangle containing both (two edges sharing a vertex fix the third vertex; disjoint edges lie in no common triangle). Hence in $\boldsymbol{A}(\boldsymbol{w}) = \sum_\sigma w_\sigma \boldsymbol{b}_\sigma \boldsymbol{b}_\sigma^\top$, the (e, e') entry equals

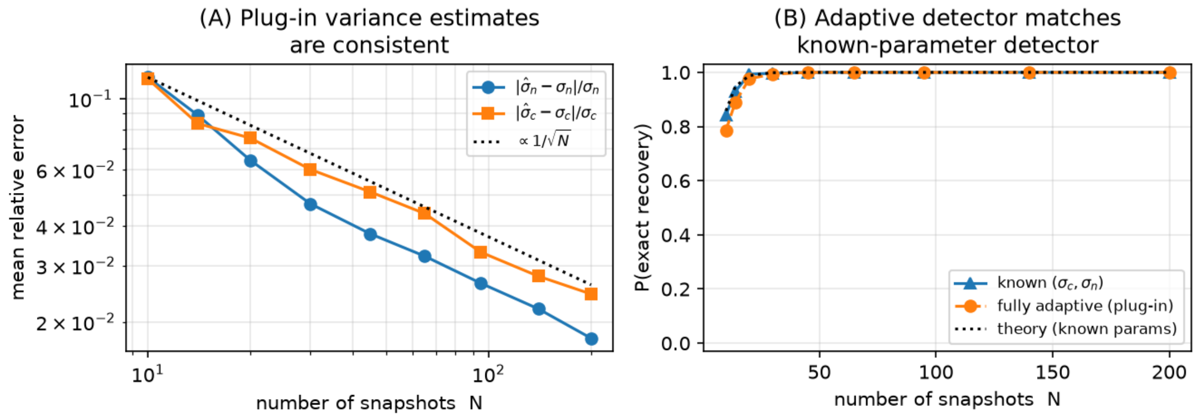

Figure 3: Plug-in estimation on the strip benchmark of the main paper's Fig. 2 ($\rho = 8$, 4/9 active). (A) Consistency of (4)–(5) (after refinement). (B) The fully adaptive detector matches the known-parameter detector. Script: `experiments/run_plugin.py`.

$w_\tau\, \boldsymbol{b}_\tau(e)\boldsymbol{b}_\tau(e') = \pm w_\tau$: no other candidate contributes. $\boldsymbol{A}(\boldsymbol{w}) = \boldsymbol{0}$ therefore forces $w_\tau = 0$ for every τ. $\square$

Corollary S4.2 (Second-order identifiability)**.** *Under the main paper's model (2), the map* $\mathcal{S} \mapsto \mathrm{Cov}(\boldsymbol{f}) = \sigma_g^2 \boldsymbol{B}_1^\top \boldsymbol{B}_1 + \sigma_c^2 \sum_{\tau\in\mathcal{S}} \boldsymbol{b}_\tau \boldsymbol{b}_\tau^\top + \Sigma_{\text{harm}} + \sigma_n^2 \boldsymbol{I}$ *is injective in* $\mathcal{S}$*; equivalently, in the curl coordinate* $\boldsymbol{z} = \boldsymbol{Q}^\top \boldsymbol{f}$ *(*$\boldsymbol{Q}$ *an orthonormal basis of* $\mathrm{im}\, \boldsymbol{B}_2$, $r = \mathrm{rank}\, \boldsymbol{B}_2$*),*

$$\boldsymbol{\Sigma}_z(\mathcal{S}) = \sigma_n^2 \boldsymbol{I}_r + \sigma_c^2 \sum_{\tau\in\mathcal{S}} \boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top, \qquad \boldsymbol{u}_\tau = \boldsymbol{Q}^\top \boldsymbol{b}_\tau, \tag{6}$$

determines $\mathcal{S}$ *uniquely. (The gradient and harmonic covariance terms are annihilated by* $\boldsymbol{Q}^\top$*; since* $\boldsymbol{b}_\tau \in \mathrm{im}\, \boldsymbol{B}_2$*, the compression to curl coordinates loses nothing:* $\boldsymbol{u}_\sigma^\top \boldsymbol{u}_\tau = \boldsymbol{b}_\sigma^\top \boldsymbol{b}_\tau = G_{\sigma\tau}$*.)*

S4.2 First-order impossibility

Proposition S4.3 (Deterministic indistinguishability)**.** *If* $\mathrm{im}\, \boldsymbol{B}_{2,\mathcal{S}} = \mathrm{im}\, \boldsymbol{B}_{2,\mathcal{S}'}$ *and the triangle signals are unknown deterministic nuisance parameters, the families of flow distributions* $\{\mathcal{N}(\boldsymbol{B}_{2,\mathcal{S}}\boldsymbol{y} + \boldsymbol{m},\ \sigma_n^2 \boldsymbol{I}) : \boldsymbol{y}\}$ *and* $\{\mathcal{N}(\boldsymbol{B}_{2,\mathcal{S}'}\boldsymbol{y}' + \boldsymbol{m},\ \sigma_n^2 \boldsymbol{I}) : \boldsymbol{y}'\}$ *coincide for every nuisance mean* $\boldsymbol{m}$*; consequently no test distinguishes* $\mathcal{S}$ *from* $\mathcal{S}'$ *at any SNR and any* N*.*

Proof. Equal column spaces: for every $\boldsymbol{y}$ there is $\boldsymbol{y}'$ with $\boldsymbol{B}_{2,\mathcal{S}'}\boldsymbol{y}' = \boldsymbol{B}_{2,\mathcal{S}}\boldsymbol{y}$ and vice versa; apply per snapshot. $\square$

Tetrahedral confusers realize the hypothesis: the four faces of a tetrahedron satisfy $\boldsymbol{b}_{012} - \boldsymbol{b}_{013} + \boldsymbol{b}_{023} - \boldsymbol{b}_{123} = \boldsymbol{0}$ (boundary of the solid 3-simplex), so any three faces span the same 3-space as any other three. On K_n ($n \geq 4$) there are $\binom{n}{4}$ tetrahedra.

S4.3 The price: exact separation and Chernoff exponent

Lemma S4.4 (Universal swap separation)**.** *Let* $\mathcal{S}' = \mathcal{S}\setminus\{\tau_a\}\cup\{\tau_b\}$ *with* τ_a, τ_b *faces of a common tetrahedron. Then* $\boldsymbol{\Sigma}_z(\mathcal{S}) - \boldsymbol{\Sigma}_z(\mathcal{S}') = \sigma_c^2(\boldsymbol{u}_a \boldsymbol{u}_a^\top - \boldsymbol{u}_b \boldsymbol{u}_b^\top)$ *and*

$$\|\boldsymbol{u}_a \boldsymbol{u}_a^\top - \boldsymbol{u}_b \boldsymbol{u}_b^\top\|_F^2 = \|\boldsymbol{u}_a\|^4 + \|\boldsymbol{u}_b\|^4 - 2(\boldsymbol{u}_a^\top \boldsymbol{u}_b)^2 = 9 + 9 - 2 = 16,$$

independent of n, of $|\mathcal{S}|$, and of the hosting tetrahedron (any two faces share exactly one edge, so $\boldsymbol{u}_a^\top \boldsymbol{u}_b = G_{ab} = \pm 1$).

Proposition S4.5 (Chernoff exponent of the minimal confuser)**.** *Let C_G be the Chernoff information between $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_z(\mathcal{S}))$ and $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_z(\mathcal{S}'))$ for a tetrahedral swap. As $\rho_2 \to 0$,*

$$C_G = \tfrac{1}{16}\, \rho_2^2\, \|\boldsymbol{u}_a \boldsymbol{u}_a^\top - \boldsymbol{u}_b \boldsymbol{u}_b^\top\|_F^2\, (1 + o(1)) = \rho_2^2\, (1 + o(1)),$$

and the optimal error of the binary test from N snapshots satisfies $P_{\text{err}} \le e^{-NC_G}$, with matching exponent ($\liminf_N -\frac{1}{N} \log P_{\text{err}} = C_G$). Hence $N^\star(\delta) \sim \log(1/\delta)/\rho_2^2$.

Proof sketch. Write $\boldsymbol{E} = \boldsymbol{\Sigma}_0^{-1/2}(\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_0)\boldsymbol{\Sigma}_0^{-1/2}$. The Chernoff exponent between equal-mean Gaussians is $\max_s \frac{1}{2}[\log\det((1-s)\boldsymbol{W}_0 + s\boldsymbol{W}_1) - (1-s)\log\det \boldsymbol{W}_0 - s \log\det \boldsymbol{W}_1]$; a second-order expansion in $\boldsymbol{E}$ around $\boldsymbol{E} = \mathbf{0}$ is maximized at $s = 1/2$ (Bhattacharyya) with value $\frac{1}{16}\operatorname{tr}\boldsymbol{E}^2 + O(\|\boldsymbol{E}\|^3)$. Here $\boldsymbol{E} = \rho_2(\boldsymbol{u}_a\boldsymbol{u}_a^\top - \boldsymbol{u}_b\boldsymbol{u}_b^\top) + O(\rho_2^2)$ (the shared atoms of $\mathcal{S} \cap \mathcal{S}'$ perturb $\boldsymbol{\Sigma}_0^{-1/2}$ only at $O(\rho_2)$, entering C_G at $O(\rho_2^3)$), so $\operatorname{tr}\boldsymbol{E}^2 = \rho_2^2 \cdot 16\,(1 + O(\rho_2))$ by Lemma S4.4. The bound $P_{\text{err}} \le e^{-NC_G}$ is the Chernoff bound; exponent tightness is classical for i.i.d. observations. $\square$

Proposition S4.6 (Fano over the confuser family)**.** *On K_n, consider the $M = 4\binom{n}{4}$ hypotheses obtained by choosing a tetrahedron and leaving one of its four faces out of an otherwise-active tetrad. Any estimator with error $\le \epsilon$ over the uniform prior needs*

$$N \;\ge\; \frac{(1-\epsilon)\log\!\big(4\binom{n}{4}\big) - \log 2}{\max_{pairs} \mathrm{KL}\big(\boldsymbol{\Sigma}_z \| \boldsymbol{\Sigma}_z'\big)} \;\asymp\; \frac{\log n}{\rho_2^2},$$

i.e. full support recovery over the confuser family pays a $\log n$ factor on top of the pairwise budget.

S4.4 Numerical verification (released tests)

`tests/test_second_order.py` certifies, deterministically or by Monte-Carlo: (i) Lemma S4.1 on K_4–K_9, strips, disjoint unions, and random clique complexes; (ii) injectivity of $\mathcal{S} \mapsto \boldsymbol{\Sigma}_z(\mathcal{S})$ *exhaustively* over all 2^4 supports of K_4; (iii) the deterministic replication of first-order confusers (residual $< 10^{-9}$); (iv) separation exactly 16 for every tetrahedral swap, and (exhaustively on K_5, all support sizes $k \le 5$) minimality of 16 over *all* equal-image pairs; (v) $C_G/\rho_2^2 \in [0.95, 1.005]$ for $\rho_2 \le 10^{-2}$ and the Monte-Carlo error of the optimal test within the e^{-NC_G} envelope with empirical exponent decreasing onto C_G; (vi) exact greedy covariance-atom recovery on rank-deficient K_5 (the regime where first-order arguments forbid it).

S5 The cyclone study: construction details

This section records the methodological details of the main paper's real-data recovery experiment (Fig. 3 there) beyond what fits in the page budget.

From winds to edge flows. ERA5 10 m winds (ARCO-ERA5 public archive, 0.703° equiangular grid, 6-hourly), Western North Pacific (0–45°N, 100–180°E), Aug–Sep 2020 (244 snapshots; the season contains 13 IBTrACS storms, incl. Bavi, Maysak, Haishen). Mesh nodes subsample every third grid point ($\approx 2.1°$); each grid cell splits into two right triangles; edge flow = trapezoidal line integral $\frac{1}{2}(\boldsymbol{V}_i + \boldsymbol{V}_j)\cdot(\Delta x, \Delta y)$ in the local equirectangular metric ($\Delta x = R\cos(\bar{\phi})\Delta\lambda$).

The mesh is simply connected, so #triangles $= |\mathcal{E}| - |\mathcal{V}| + 1 = \dim\ker \boldsymbol{B}_1$ and $\boldsymbol{B}_2$ has full column rank (verified: 1554/1554) — the favorable-geometry regime of the dichotomy, with no equal-image confusers. By Stokes' theorem the curl statistic of a triangle approximates its circulation, i.e. area-integrated relative vorticity; a sign convention (sorted-vertex traversal vs. geometric orientation) is tracked explicitly in the code and is irrelevant to the energy detector.

Detection. Per 4-day window (16 snapshots), the temporally centered decorrelated scores $\hat{\boldsymbol{y}} = \boldsymbol{G}^{-1}\boldsymbol{c}$ are computed with NO oracle parameters: centering removes any window-constant background exactly, and thresholds/scales are plug-in (§S3 machinery). Windows are the device that maps the i.i.d.-snapshot theory onto moving weather: a cyclone translates $\sim$3–6°/day, so over 4 days it crosses a few 2.1° triangles — the "support" is quasi-static at the window scale, and the per-window scores are pooled across the season for evaluation. This is an idealization and is labeled as such (README honesty note 9).

Validation. Internal: relative vorticity $\zeta = \partial_x v - \partial_y u$ by centered differences on the FULL 0.7° grid (the detector only ever sees the 2.1° mesh flows); per-triangle truth = window-mean $|\zeta|$ averaged over grid points inside the triangle; ROC sweeps the detector score against truth binarized at typhoon-scale $3 \times 10^{-5}\,\mathrm{s}^{-1}$. External: IBTrACS v04r01 best-track fixes ($\geq 34\,\mathrm{kt}$ where wind is reported), a triangle labeled positive if a fix falls within 1.5° of its centroid during the window. The external reference shares nothing with ERA5's assimilation of winds into our statistic — it is the community's independent record of where cyclones actually were. A synthetic Rankine-vortex test (`tests/test_geo.py`) verifies the whole bridge end-to-end: localization at the core, correct cyclonic sign after orientation correction, and rank agreement with the finite-difference vorticity.

S6 Reproducibility

All figures regenerate on CPU from fixed seeds via `experiments/run_all.py`; individually: Fig. 1 $\leftarrow$ `run_fano.py` (pure theory, seconds); Fig. 2 $\leftarrow$ `run_partial_sampling.py` ($\approx$4 min); Fig. 3 $\leftarrow$ `run_plugin.py` ($\approx$3 min); the main paper's cyclone study $\leftarrow$ `run_real_cyclone.py` ($\approx$3 min from the cached ERA5/IBTrACS files). The test-suite (`pytest -q`, 34 tests) checks: the Fano bounds against the empirical recovery boundary of the main paper's Fig. 1 (a converse must lower bound it), the closed form (3) against Monte-Carlo, the envelope of Lemma S3.1, the adaptive-vs-known recovery gap, every claim of Section S4 (spark, exhaustive K_4 injectivity, separation 16, $C_G/\rho_2^2 \to 1$, Chernoff envelope, rank-deficient recovery), and the wind-to-edge-flow bridge (mesh full rank by Euler; a synthetic Rankine vortex localizes at the correct triangles with the correct sign and matches the finite-difference vorticity ranking).

References

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